Sentiment Analysis of Football-Related Tweets: A Comparative Study Between Cristiano Ronaldo and Lionel Messi Fan Discourse

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Abstract

This study presents a complete pipeline for sentiment analysis applied to a large-scale corpus of English-language tweets discussing two prominent football players, Cristiano Ronaldo and Lionel Messi. The dataset, sourced from Kaggle, comprises over one million tweets, from which a representative sample of 100,000 entries was selected for computational efficiency. Text preprocessing was performed using a combination of regular expression cleaning, spaCy-based lemmatization, and stopword removal. Sentiment labels were automatically assigned using the VADER lexicon, producing three classes: positive, neutral, and negative. Tweets were transformed into numerical features using TF-IDF vectorization with bigram support. Four classical machine learning classifiers were evaluated: Support Vector Machine (SVM), Multinomial Naive Bayes, Logistic Regression, and Random Forest. Among them, SVM achieved the best performance, attaining a test accuracy of 80.46% and a macro-averaged F1-score of 0.7914 on the held-out test set, with a 5-fold cross-validation accuracy of 0.7962. Results indicate that fan discourse about both players is predominantly positive, with vocabulary and thematic focus remaining highly similar across both groups.

1 Introduction

The rapid proliferation of social media platforms, particularly microblogging services such as Twitter (now X), has produced an unprecedented volume of user-generated textual content. Sports, and football in particular, constitute one of the most discussed topics online, generating millions of opinionated messages daily. Analysing the sentiment embedded in such content offers valuable insights into public perception, fan engagement, and player popularity.

This work addresses the problem of automatic sentiment classification in sports-related social media text, framing it as a three-class supervised learning task (positive, neutral, negative). We focus on tweets mentioning Cristiano Ronaldo and Lionel Messi, arguably the two most discussed athletes of the modern era, which offers a natural basis for comparative analysis.

The contributions of this study are threefold: (1) a reproducible preprocessing pipeline tailored to noisy social media text; (2) a systematic comparison of four classical machine learning approaches under rigorous cross-validation; and (3) an exploratory analysis of sentiment dynamics, including temporal trends and player-level differences.

2 Background

Sentiment analysis, also referred to as opinion mining, is a well-established subfield of Natural Language Processing (NLP) concerned with identifying and extracting subjective information from text [1]. Its application to social media data presents unique challenges, including informal language, code-switching, and heavy use of irony or sarcasm.

Lexicon-based methods, such as VADER (Valence Aware Dictionary and sEntiment Reasoner), were specifically designed to handle short social media texts. Mohammad and Turney [2] demonstrated that crowdsourced methods can produce reliable sentiment and emotion lexicons with high inter-annotator agreement, supporting rule-based approaches as competitive baselines.

The use of TF-IDF combined with linear classifiers—particularly SVM—remains a strong baseline for text classification tasks, as established by Joachims [3]. More recent work by Go et al. [4] applied distant supervision using emoticons to label Twitter data, reporting accuracies around 80% on binary sentiment classification tasks, a benchmark broadly consistent with our findings.

Deep learning approaches, including LSTM and Transformer-based architectures such as BERT [5], have since surpassed classical methods on many NLP benchmarks. However, they incur substantially higher computational costs and typically require larger labelled datasets. For settings where interpretability, efficiency, or limited data are prioritised, classical pipelines remain competitive.

3 Methodology

3.1 Data Acquisition and Filtering

The dataset was obtained via KaggleHub from the repository *Twitter Sentiment Analysis and Word Embeddings* (Ibrahim Serouis). Raw CSV files for both players were loaded into pandas DataFrames. The following sequential filtering steps were applied:

1. Retention of English-language tweets only (`lang == "en"`).

2. Deduplication based on tweet content.

3. Removal of tweets with fewer than 20 characters.

A stratified random sample of 100,000 tweets was subsequently drawn for computational tractability.

3.2 Text Preprocessing

A two-stage cleaning pipeline was implemented. First, basic cleaning normalised the raw text: conversion to lowercase, removal of URLs, `@mentions`, and non-alphabetic characters. Second, advanced linguistic preprocessing was applied via spaCy (`en_core_web_sm`), disabling the dependency parser and named entity recognition to reduce latency. This stage performed tokenization, lemmatization, and removal of stopwords, punctuation, and tokens of two characters or fewer. Contraction artefacts (*don*, *ve*, *ll*) were explicitly filtered.

3.3 Sentiment Labelling

Ground-truth labels were generated automatically using NLTK's SentimentIntensityAnalyzer (VADER). Each tweet received a compound polarity score in $[-1, 1]$, subsequently discretised as follows:

$$\text{label} = \begin{cases} \textit{positive} & \text{if } s > 0.05 \\ \textit{negative} & \text{if } s < -0.05 \\ \textit{neutral} & \text{otherwise} \end{cases}$$

3.4 Feature Extraction

Preprocessed texts were transformed into a sparse TF-IDF matrix using `TfidfVectorizer` with the following configuration: vocabulary limited to 5,000 features; n-gram range of $(1, 2)$ to capture phrasing; English stopword removal.

3.5 Classifiers

Four classifiers were trained: Linear SVM (`LinearSVC`, `class_weight=balanced`, `max_iter=3000`), Multinomial Naive Bayes, Logistic Regression (`class_weight=balanced`, `max_iter=1000`), and Random Forest (100 estimators, parallelised). An 80/20 stratified train-test split was applied, and model selection was performed via 5-fold stratified cross-validation on the training partition.

4 Experimentation

4.1 Dataset Characteristics

After all preprocessing and filtering, the working sample comprised 100,000 tweets (Table 1). The player distribution is slightly imbalanced after filtering, with Ronaldo-related tweets representing a larger proportion, though not severely so.

Table 1: Dataset summary statistics.

Property	**Value**
Source	Kaggle (Serouis, 2023)
Original size	>1,000,000 tweets
Working sample	100,000 tweets
Language	English only
Classes	Positive / Neutral / Negative
Labelling method	VADER (automated)
TF-IDF features	5,000
n-gram range	(1, 2)
Train / Test split	80% / 20%

4.2 Evaluation Metrics

All models were evaluated using: (1) accuracy on the held-out test set; (2) macro-averaged F1-score, which weights each class equally regardless of support; (3) weighted F1-score; and (4) mean cross-validation accuracy $\pm$ standard deviation across 5 folds.

4.3 Implementation Details

Experiments were conducted in Google Colab using Python 3.10, scikit-learn 1.x, spaCy 3.x, and NLTK 3.x. Processing of 100,000 tweets through the spaCy pipeline took approximately 8–12 minutes on a standard CPU runtime. No GPU was used. All random seeds were fixed to 16 for reproducibility.

5 Results and Analysis

5.1 Sentiment Distribution

The sentiment analysis reveals a clear dominance of positive discourse in the football-related Twitter sphere. As illustrated in Figure 1, the absolute volume of tweets is higher for Ronaldo across all categories, which is consistent with the larger sample size obtained for this player post-filtering. However, when observing the relative proportions in Table 2, we find a remarkable structural similarity between the two fan bases.

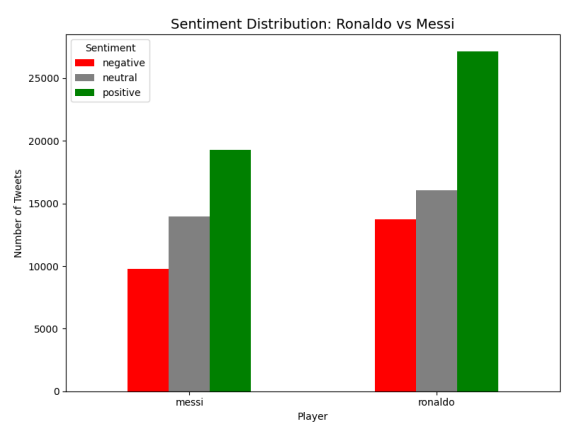

Figure 1: Absolute distribution of sentiment classes for Ronaldo and Messi.

Table 2: Proportional Sentiment Distribution by Player

Player	Positive	Neutral	Negative
Ronaldo	0.48	0.28	0.24
Messi	0.45	0.32	0.23

The disaggregated data shows that positive sentiment exceeds 40% for both players (48% for Ronaldo and 45% for Messi), significantly higher than the negative sentiment (24% for Ronaldo and 23% for Messi). The mean compound VADER scores are also positive, with Ronaldo scoring marginally higher.

Temporal analysis, visualized in Figure 2, indicates that this positive trend is stable throughout the period, though Ronaldo exhibits higher short-term volatility. This is evidenced by the more frequent and extreme fluctuations in his average sentiment score, possibly reflecting a more polarized or reactive fan base during specific match events or news cycles.

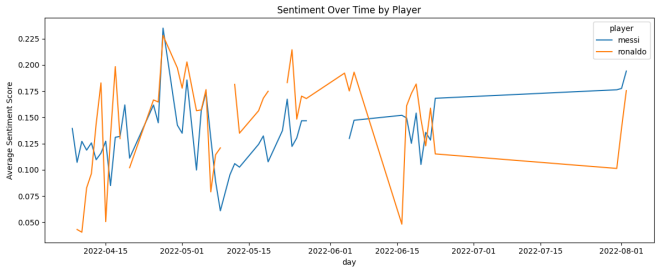

Figure 2: Average sentiment score over time for Messi and Ronaldo.

5.2 Classifier Performance

Table 3 summarises quantitative results for all four models. SVM achieves the best performance on both validation and test metrics. Notably, the small gap between cross-validation and test accuracy across all models indicates low overfitting risk, reflecting the generalisability of TF-IDF-based linear classifiers for this type of task.

Table 3: Model comparison: cross-validation and test set performance.

Model	CV Acc.	CV Std	Test Acc.	Macro F1
SVM	**0.7962**	±0.0017	**0.8046**	**0.7914**
Log. Regression	0.7874	±0.0028	0.7956	0.7850
Random Forest	0.7615	±0.0032	0.7693	0.7392
Naive Bayes	0.6991	±0.0015	0.7059	0.6775

5.3 Error Analysis

The confusion matrix for SVM (Figure 3) reveals that most misclassifications occur between adjacent sentiment classes (negative↔neutral and neutral↔positive). This is consistent with the inherent ambiguity of sentiment boundaries in natural language. The negative class exhibits the lowest recall, indicating that clearly negative tweets are the hardest to identify, a known challenge for lexicon-labelled social media data.

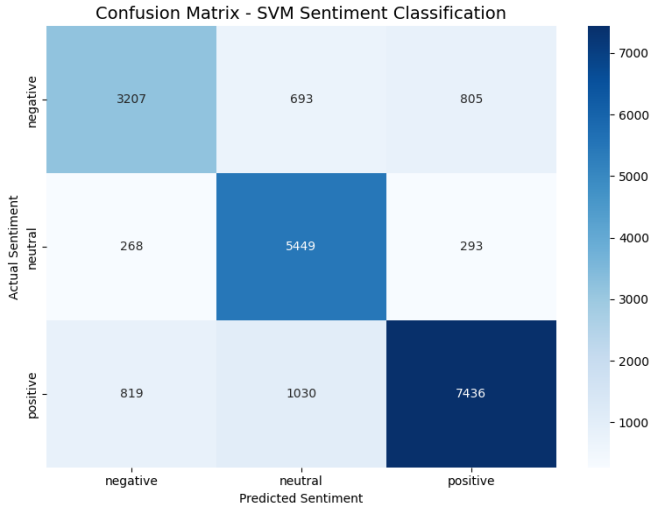

Figure 3: Confusion Matrix for the Linear SVM classifier.

5.4 Lexical Insights

Word frequency analysis, visualized in Figure 4, confirms that both player-specific corpora are dominated by a shared footballing lexicon. Terms such as *goal*, *player*, *team*, and *season* appear with high central prominence in both clouds, suggesting that despite the rivalry, the thematic structure of the discourse remains remarkably consistent across both fan bases. Furthermore, player names appear frequently cross-referenced, indicating a highly comparative discourse among fans.

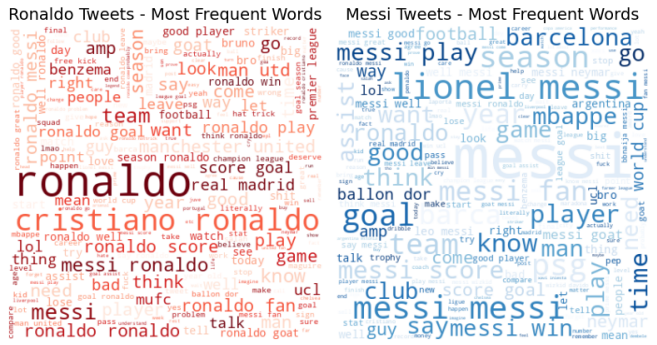

Figure 4: Comparative word clouds for Ronaldo (left) and Messi (right) tweet corpora, illustrating the high overlap in thematic focus and vocabulary.

Sentiment-stratified vocabulary shows minor but logical

differentiation: positive tweets are enriched with terms such as *good*, *win*, and *well*, while negative tweets are characterized by performance-related criticism and terms like *bad*.

Average engagement metrics disaggregated by sentiment (Table 4) indicate that neutral and positive tweets receive higher interaction than negative ones. This suggests that community reinforcement on Twitter is stronger for constructive or celebratory content than for negative discourse in the context of football fandom.

Table 4: Engagement Metrics by Sentiment

Sentiment	**Avg. Likes**	**Avg. Retweets**
Positive	32.00	2.75
Neutral	33.10	2.59
Negative	15.79	1.34

6 Conclusions

This project presented a complete pipeline for sentiment analysis on social media data, focusing on tweets related to Cristiano Ronaldo and Lionel Messi. Starting from raw data acquisition, exploratory data analysis was performed and several preprocessing techniques were applied to clean and normalise the textual content. Using NLP methods such as tokenization and lemmatization, text was transformed into a structured format suitable for machine learning. TF-IDF vectorization was then used for feature extraction, and four classifiers were trained and evaluated.

Among them, SVM achieved the best performance, reaching a test accuracy of 80.46% and strong F1-scores across all sentiment classes, demonstrating that linear models combined with TF-IDF features are highly effective for text classification in high-dimensional sparse spaces. From an analytical perspective, the study revealed that: (1) social media discussions about both players are predominantly positive; (2) vocabulary and topics are highly similar across both groups; and (3) differences in sentiment between players are present but relatively small. Overall, the results highlight the effectiveness of classical NLP and machine learning techniques for sentiment analysis, while also emphasising the challenges posed by noisy and context-dependent social media data.

Limitations. Despite these promising results, several limitations should be acknowledged. First, sentiment labels were automatically generated using VADER, a lexicon-based approach. While efficient and well-suited for social media, this introduces a dependency between the training data and the labelling method, meaning that the supervised models effectively learn to approximate VADER rather than human-annotated sentiment. Additionally, lexicon-based methods may fail to capture complex linguistic phenomena such as sarcasm, irony, or context-dependent meaning, introducing noise into the ground truth. Second, the TF-IDF representation, although effective, does not capture semantic relationships between words, limiting the model's ability to understand context and nuanced meanings. Finally, the analysis was conducted on a sampled subset of the data (100,000 tweets) for computational efficiency, which may reduce exposure to less frequent linguistic patterns present in the full dataset.

Future work. Several directions could address these limitations. First, the adoption of deep learning architectures such as LSTM networks or Transformer-based models (e.g., BERT, RoBERTa) would better capture contextual and long-range dependencies in text. Second, incorporating static or contextualised word embeddings (Word2Vec, GloVe, FastText) would enhance semantic representation beyond bag-of-words approaches. Third, fine-tuning pretrained language models specifically for social media sentiment tasks could yield significant accuracy gains. Additionally, enriching the feature space with user metadata and engagement metrics, and addressing class imbalance more robustly through data augmentation or semi-supervised learning, represent promising avenues for future research.

References

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